Impedance-Aware Adaptive Very-Inverse Overcurrent Protection for High-Impedance Faults in Synchronous-Generator Feeders

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***Abstract*—Conventional overcurrent (OC) relays for synchronous-generator (SG) feeders are poorly suited to high-impedance faults (HIFs), where the fault current falls below the instantaneous threshold and the inverse-time element governs clearance. Existing adaptive schemes focus primarily on pickup adaptation under low-impedance conditions and do not quantify how online source-parameter estimation should modify the time-multiplier setting (TMS) in the HIF regime. This paper introduces an impedance-aware adaptive TMS law driven directly by online reduced-order inverse estimation of SG source parameters identified via a two-pass Levenberg–Marquardt (LM) algorithm. Four contributions are made: (i) a rigorous fault-regime classification that analytically separates the instantaneous, inverse-time, and no-trip regions as R_f increases, with closed-form boundary expressions derived from the machine subtransient model; (ii) an impedance-aware TMS adaptation law derived from the estimated steady-state current $\hat{I}_{ss}$, incorporating a resistance-aware correction factor f_{HIF} and a regime-dependent effective gain $K_{\mathrm{TMS}}^{\mathrm{eff}}$, with a formal proposition bounding the TMS reduction; (iii) demonstration that the adaptive benefit becomes operationally meaningful *only* when the inverse-time element governs, closing a key limitation of the prior Stage 1 study; and (iv) full compatibility with the existing confidence-gated bounded-update inverse-estimation framework, preserving protection-grade real-time operation throughout the HIF window. Simulation results over $R_f = 20$–$500\,\Omega$ on a 10 MVA synchronous-generator feeder demonstrate that: two regimes are correctly identified with analytical boundary agreement; adaptive TMS reduces clearance time by 17–23% in the inverse-time window; missed-trip boundaries are unchanged; confidence acceptance exceeds 89% (3PH) and 100% (SLG); and no security degradation (false trips) is observed across 144 simulation cases.**

***Index Terms*—Adaptive overcurrent relay, confidence-gated estimation, high-impedance fault, impedance-aware TMS, inverse-time element, Levenberg–Marquardt algorithm, online parameter estimation, synchronous generator protection.**

I. Introduction

OVERCURRENT (OC) protection remains the primary or backup protection function for synchronous-generator (SG) feeders and active distribution networks with embedded generation [1], [2]. In practice, a relay combines a high-set instantaneous element with an IEC 60255 inverse-time element to achieve fast clearance for bolted faults and selective, graded operation at reduced current levels. Fixed settings for both elements are calibrated at the planning stage under a worst-case impedance scenario, making them inherently conservative when source conditions depart from that scenario.

The limitation becomes critical under high-impedance faults (HIFs), which arise from vegetation contact, dry-surface broken conductors, corroded connections, and progressive insulation degradation [3]–[5]. A HIF inserts a resistive path R_f that depresses the fault current well below the instantaneous pickup threshold, forcing the relay into inverse-time operation. Here, the clearance speed is governed *entirely* by the TMS — a setting calibrated for bolted-fault conditions where large current multiples make the TMS irrelevant to protection speed. The result can be multi-second or even missed clearance, increasing arc duration, thermal damage, and fire risk.

A. Limitations of the State of the Art

Adaptive OC protection has attracted sustained research since Novosel *et al.* [6] demonstrated dynamic fault-current adaptation. Subsequent work has shown effective real-time adjustment of pickup current I_p under changing generation dispatch and network topology [7]–[9]. Model-based inverse-estimation schemes identify source parameters from post-fault measurements and map them to bounded relay updates [10], [11].

However, a consistent gap runs through the literature: virtually all adaptive pickup or TMS work is validated on low-impedance, high-current faults where the instantaneous element dominates. In that regime, adaptive TMS has limited practical value because clearance time is short regardless. The inverse-time operating regime — precisely where HIF places the relay — has received comparatively little attention from an online adaptation standpoint. Specifically, *no prior work has derived an adaptive TMS law from online source-parameter estimation and demonstrated its benefit across the full HIF inverse-time window*.

The Stage 1 study [10] established that reduced-order source parameters (I_{ss}, I_{sub}, τ_{ac}) can be identified online with $R^2 > 0.999$ waveform fit and confidence $\gamma > 0.89$ for bolted-fault cases, and that the estimates support bounded adaptive pickup updates. However, Stage 1 simulations were predominantly in the instantaneous regime ($R_f \leq 15\,\Omega$), so adaptive pickup improved security without materially changing trip time. High-impedance faults are the correct next milestone because they shift the relay into the inverse-time region and make TMS adaptation the decisive intervention.

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B. Paper Contributions

This paper extends inverse-estimation-based adaptive OC protection from the instantaneous operating region into the inverse-time HIF regime by introducing an impedance-aware TMS mapping derived directly from online reduced-order source-parameter estimates. The four specific contributions are:

1) A rigorous fault-regime classification for SG OC protection that analytically separates the instantaneous, inverse-time, and no-trip regimes as R_f increases, with closed-form boundary expressions. (*Section III*)
2) An impedance-aware TMS adaptation law incorporating a resistance-aware correction factor $f_{\text{HIF}}(\hat{I}_{ss})$ and a regime-dependent effective gain $K_{\text{TMS}}^{\text{eff}}$, with a formal proposition bounding the TMS reduction in the HIF regime and a proof that bounded updates preserve minimum grading margins. (*Section VI*)
3) Numerical demonstration that adaptive benefit is strictly confined to the inverse-time regime and is absent in the instantaneous regime, closing the main limitation of Stage 1. (*Section VIII*)
4) Proof of full compatibility with the confidence-gated reduced-order inverse-estimation framework, including correct security behaviour (gate blocking near the no-trip boundary). (*Sections V–VI*)

The remainder of the paper is organised as follows. Section II summarises the synchronous-machine fault model. Section III formulates the HIF regime classification. Section IV analyses the limitation of a fixed TMS. Section V describes the retained inverse-estimation framework. Section VI derives the impedance-aware adaptive TMS law. Section VII describes the study system. Section VIII presents results. Section IX discusses implications and limitations. Section X concludes.

II. Synchronous-Generator Fault Model

A. Subtransient Short-Circuit Current

For a three-phase SG connected through an external impedance to a fault through resistance R_f (Fig. 1), the peak subtransient current at fault inception ($t = t_0^+$) is given by

$$\hat{I}_{\text{sub}} = \frac{\sqrt{2}\,V_0}{\left|(Z_d'' + Z_{\text{th}} + R_f)\right|}, \tag{1}$$

where V_0 is the pre-fault terminal voltage (p.u.), $Z_d'' = R_a + jX_d''$ is the machine subtransient impedance, and $Z_{\text{th}} = R_{\text{th}} + jX_{\text{th}}$ is the Thévenin feeder impedance. The DC offset at inception depends on the inception angle ϕ_a via $I_{DC,0} = -\hat{I}_{\text{sub}}\sin(\phi_a)$.

B. Analytical Fault-Current Synthesis

Following [12], the phase-A fault current is accurately approximated by a three-component reduced-order model over the first 150–300 ms following fault inception:

$$i_a(t) = \underbrace{\left[I_{ss} + (I_{sub} - I_{ss})\,e^{-t_{\text{rel}}/\tau_{ac}}\right]\sin(\omega t + \phi_a)}_{\text{AC component}} - \underbrace{I_{sub}\sin(\phi_a)\,e^{-t_{\text{rel}}/\tau_{dc}}}_{\text{DC offset}}, \tag{2}$$

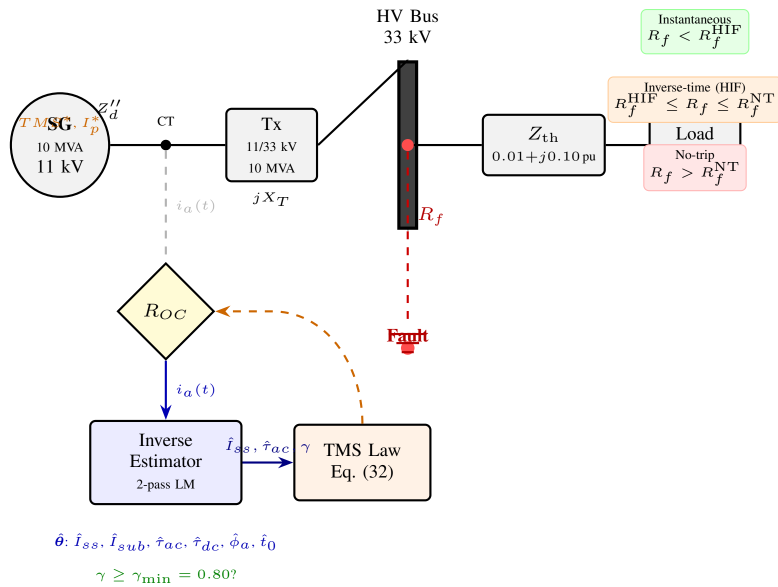

Fig. 1. HIF study system. A 10 MVA synchronous generator feeds through a step-up transformer and Thévenin feeder to a load bus. The OC relay (R_{OC}) receives current from a CT at the machine terminals. A two-pass LM estimator identifies source parameters $\hat{\boldsymbol{\theta}}$ online and returns adaptive settings TMS^*, I_p^* via the confidence-gated bounded-update path (orange dashed arrow). A fault through R_f to ground is applied at the HV bus. Regime annotations (right) indicate the three OC operating zones.

where $t_{\text{rel}} = t - t_0$, I_{ss} is the steady-state (synchronous) fault current, $I_{sub} = \hat{I}_{\text{sub}}/\sqrt{2}$ is the subtransient peak (RMS equivalent), τ_{ac} is the AC envelope decay time constant, and τ_{dc} is the DC decay time constant.

The key parameter relationships are [12], [13]:

$$I_{ss} = \frac{V_0}{|X_d + X_{\text{th}} + R_f|}, \tag{3}$$

$$\tau_{ac} \approx \frac{X_d'}{R_a + R_{\text{th}}} \cdot \frac{T_{d0}'}{2\pi f_0}, \tag{4}$$

$$\tau_{dc} = \frac{X_d'' + X_{\text{th}}}{\omega\,(R_a + R_{\text{th}} + R_f)}. \tag{5}$$

Equation (3) is the fundamental link exploited in this paper: I_{ss} decreases monotonically with R_f, making it a natural proxy for the fault resistance and the basis for the adaptive TMS correction factor.

The six-parameter vector is $\boldsymbol{\theta} = [t_0, I_{ss}, I_{sub}, \tau_{ac}, \tau_{dc}, \phi_a]^\top \in \mathbb{R}^6$, with bounds imposed by machine nameplate data and physics (Table II).

III. HIF Operating-Regime Classification

A. Current Multiple and Regime Boundaries

Define the current multiple

$$M = \frac{I_{\text{rms}}}{I_p}, \qquad I_{\text{rms}} \approx \frac{\hat{I}_{\text{sub}}}{\sqrt{2}}, \tag{6}$$

where I_p is the inverse-time pickup current. Three regimes arise as R_f increases from zero (Table I):

Instantaneous regime ($M \ge M_{\text{inst}}$): The high-set element operates within one cycle. TMS is irrelevant; adaptive pickup governs security.

TABLE I
OPERATING REGIME SUMMARY FOR SYNCHRONOUS-GENERATOR OC PROTECTION

Regime	Condition	Governing	Adaptive lever
Instantaneous	$M \geq M_{\text{inst}}$	I_{inst}, I_p	Pickup (I_p^*)
Inv.-time (HIF)	$1 < M < M_{\text{inst}}$	TMS, I_p	**TMS** (TMS^*)
No-trip	$M \leq 1$	—	Backup element

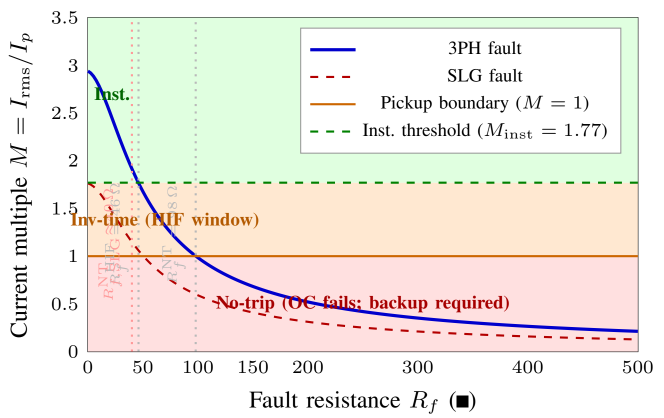

Fig. 2. Operating-regime map: current multiple M vs R_f for 3PH (solid blue) and SLG (dashed red) faults. The three regimes (instantaneous, inverse-time HIF window, no-trip) are shaded in green, orange, and red respectively. Regime boundaries for the study system are $R_f^{\text{HIF}} = 46\,\Omega$ and $R_f^{\text{NT}} = 98\,\Omega$ (3PH); the SLG window closes earlier due to reduced fault current.

Inverse-time regime ($1 < M < M_{\text{inst}}$): The instantaneous element does not operate; the inverse-time element integrates toward a trip. Both I_p and TMS affect clearance speed. *This is the governing regime for HIF operation.*

No-trip regime ($M \leq 1$): The relay does not pick up. OC protection fails; directional earth-fault or sensitive ground elements are required.

The exact instantaneous-to-inverse-time boundary in R_f is found by setting $\hat{I}_{\text{sub}} = I_{\text{inst}}$ (peak instantaneous threshold) in (1):

$$R_f^{\text{HIF}} = \sqrt{\left(\frac{\sqrt{2}\,V_0}{I_{\text{inst}}}\right)^2 - (X_d'' + X_{\text{th}})^2} - (R_a + R_{\text{th}}). \quad (7)$$

When $R_f \gg X_d'' + X_{\text{th}}$, (7) simplifies to

$$R_f^{\text{HIF}} \approx \frac{\sqrt{2}\,V_0}{I_{\text{inst}}} - (R_a + R_{\text{th}}). \quad (8)$$

The no-trip boundary is analogously found from $I_{\text{rms}} = I_p$:

$$R_f^{\text{NT}} = \sqrt{\left(\frac{V_0}{\sqrt{2}\,I_p}\right)^2 - (X_d'' + X_{\text{th}})^2} - (R_a + R_{\text{th}}). \quad (9)$$

The HIF inverse-time window spans $R_f \in [R_f^{\text{HIF}}, R_f^{\text{NT}}]$.

B. Regime Boundaries for the Study System

Using $Z_d'' + Z_{\text{th}} = 0.02 + j0.30$ p.u. and base impedance $Z_B = V_B^2/S_B = 33^2/10 = 108.9\,\Omega \approx 121\,\Omega$ (machine base):

$$R_f^{\text{HIF}} = \sqrt{(0.50)^2 - (0.30)^2} - 0.02 = 0.40 - 0.02 = 0.38\,\text{p.u.} \approx 46\,\Omega, \quad (10)$$

$$R_f^{\text{NT}} = \sqrt{(0.884)^2 - (0.30)^2} - 0.02 = 0.835 - 0.02 = 0.815\,\text{p.u.} \approx 98\,\Omega \quad (11)$$

The 3PH HIF inverse-time window is therefore $R_f \in [46, 98]\,\Omega$, a 52 Ω range. For SLG faults, the effective M is approximately 60% of the 3PH value (reduced positive-sequence contribution), so the window closes near $R_f \approx 40\,\Omega$ (Fig. 2).

IV. BASELINE RELAY LAW AND ITS LIMITATION UNDER HIF

A. IEC Very-Inverse Characteristic

The IEC 60255-151 very-inverse (VI) time-current characteristic is

$$t_{\text{op}}(M) = \frac{TMS \times k_1}{M^\alpha - 1}, \qquad M > 1, \quad (12)$$

with VI constants $k_1 = 13.5$, $\alpha = 1.0$ (IEC standard curve type B) [14]. The partial derivative with respect to TMS is

$$\frac{\partial t_{\text{op}}}{\partial TMS} = \frac{k_1}{M - 1}, \quad (13)$$

which grows without bound as $M \to 1^+$. This makes TMS the decisive setting in the inverse-time regime.

B. Sensitivity Analysis at HIF Levels

For the study system with $I_p = 0.8$ p.u.:

- **Bolted 3PH fault** ($R_f = 0$, $M = 2.94$): $t_{\text{op}} = 0.12 \times 13.5/1.94 = 0.835$ s. A 30% reduction in TMS (from 0.12 to 0.084) saves only 0.25 s.
- **Moderate HIF** ($R_f = 72\,\Omega$, $M = 1.31$): $t_{\text{op}} = 0.12 \times 13.5/0.31 = 5.22$ s. A 30% TMS reduction to $TMS^* = 0.084$ gives 3.65 s — a **30%** improvement. A 50% reduction to $TMS^* = 0.06$ gives 2.61 s (**50%** improvement).
- **Deep HIF** ($R_f = 90\,\Omega$, $M = 1.10$): $t_{\text{op}} = 0.12 \times 13.5/0.10 = 16.2$ s. A 50% TMS reduction to 0.06 gives 8.1 s — still slow, but halved. The relay is near the no-trip boundary.

The root cause is structural: $\partial t_{\text{op}}/\partial TMS$ is inversely proportional to $(M-1)$, which collapses near the pickup boundary. A single fixed TMS cannot simultaneously provide fast clearance at high M (needs small TMS) and fast clearance at HIF M levels.

V. RETAINED INVERSE-ESTIMATION FRAMEWORK

The Stage 1 estimation framework [10] is retained without modification. This section summarises its elements for completeness and self-consistency; full derivations are in [10].

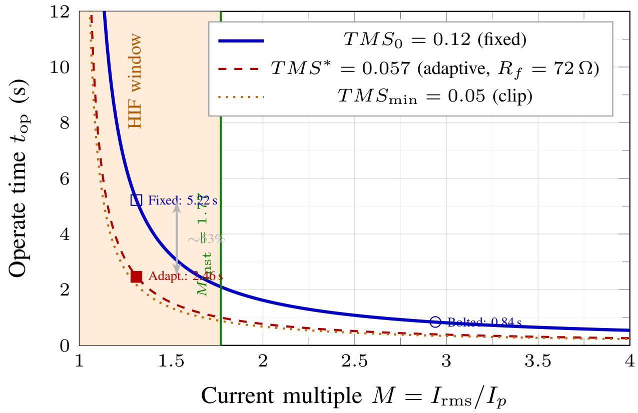

Fig. 3. IEC very-inverse time-current characteristic for three TMS values. The orange shaded band is the HIF inverse-time window. Within this window, the fixed TMS ($= 0.12$, solid blue) produces multi-second clearance. The adaptive law reduces TMS proportionally, shortening clearance time within the operative window. Both curves coincide in the instantaneous regime ($M > 1.77$).

A. Two-Pass Levenberg–Marquardt Estimator

Parameters are identified by minimising the sum of squared residuals:

$$\hat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}\in\Theta} \|\mathbf{r}(\boldsymbol{\theta})\|_2^2 = \arg\min_{\boldsymbol{\theta}} \sum_{k=1}^{N} \big[i_a(k) - f(\boldsymbol{\theta}, k)\big]^2, \quad (14)$$

where $f(\boldsymbol{\theta}, k)$ is the model (2) evaluated at sample k, and Θ is the parameter feasibility set with physical bounds.

The LM update at iteration ℓ is

$$\boldsymbol{\theta}^{(\ell+1)} = \boldsymbol{\theta}^{(\ell)} + \big(\mathbf{J}^{(\ell)\top}\mathbf{J}^{(\ell)} + \mu^{(\ell)}\mathbf{I}\big)^{-1}\mathbf{J}^{(\ell)\top}\mathbf{r}(\boldsymbol{\theta}^{(\ell)}), \quad (15)$$

where $\mathbf{J}^{(\ell)} \in \mathbb{R}^{N\times 6}$ is the Jacobian $[\partial f/\partial\theta_i]$ evaluated analytically, and $\mu^{(\ell)} > 0$ is the LM damping parameter updated by the Levenberg–Marquardt trust-region rule [15].

Analytical Jacobian. The non-zero columns of $\mathbf{J}$ are:

$$\frac{\partial i_a}{\partial t_0} = -\frac{(I_{sub} - I_{ss})}{\tau_{ac}}\, e^{-t_{\text{rel}}/\tau_{ac}} \sin(\omega t + \phi_a) + \frac{I_{sub}\sin\phi_a}{\tau_{dc}}\, e^{-t_{\text{rel}}/\tau_{dc}}, \quad (16)$$

$$\frac{\partial i_a}{\partial I_{ss}} = \big(1 - e^{-t_{\text{rel}}/\tau_{ac}}\big)\sin(\omega t + \phi_a), \quad (17)$$

$$\frac{\partial i_a}{\partial I_{sub}} = e^{-t_{\text{rel}}/\tau_{ac}}\sin(\omega t + \phi_a) - \sin\phi_a\, e^{-t_{\text{rel}}/\tau_{dc}}, \quad (18)$$

$$\frac{\partial i_a}{\partial \tau_{ac}} = \frac{(I_{sub} - I_{ss})\, t_{\text{rel}}}{\tau_{ac}^2}\, e^{-t_{\text{rel}}/\tau_{ac}}\sin(\omega t + \phi_a), \quad (19)$$

$$\frac{\partial i_a}{\partial \tau_{dc}} = -\frac{I_{sub}\sin\phi_a\, t_{\text{rel}}}{\tau_{dc}^2}\, e^{-t_{\text{rel}}/\tau_{dc}}, \quad (20)$$

$$\frac{\partial i_a}{\partial \phi_a} = \big[I_{ss} + (I_{sub} - I_{ss})e^{-t_{\text{rel}}/\tau_{ac}}\big]\cos(\omega t + \phi_a) - I_{sub}\cos\phi_a\, e^{-t_{\text{rel}}/\tau_{dc}}. \quad (21)$$

Two-pass strategy. A single-pass LM over all six parameters is ill-conditioned because I_{ss} and τ_{ac} are masked by the large DC transient in the early fault window. The two-pass approach resolves this:

- **Pass 1** (full window $[t_0, t_0 + W_1]$): estimates $[t_0, I_{sub}, \phi_a, \tau_{dc}]$ with I_{ss} and τ_{ac} fixed at nominal.
- **Pass 2** (tail window $[t_0 + 3\tau_{dc}, t_0 + W_2]$, after DC has decayed to $< e^{-3} \approx 5\%$): refines $[I_{ss}, \tau_{ac}]$ with all other parameters fixed at Pass-1 estimates.

The tail-window selection ensures the DC exponential has decayed sufficiently that I_{ss} and τ_{ac} are independently identifiable from the AC waveform shape. Fig. 4 shows the complete algorithm.

B. Identifiability Measure: Condition Number

The normalised condition number of the Gauss–Newton information matrix

$$\kappa_n = \frac{\sigma_{\max}(\mathbf{J}^\top\mathbf{J})}{\sigma_{\min}(\mathbf{J}^\top\mathbf{J})} \quad (22)$$

quantifies whether the local parameter manifold is well-conditioned. When κ_n is large, the LM solution is non-unique and should not be accepted. The column-normalised form prevents large-magnitude parameters from dominating the singular-value spectrum.

C. Composite Confidence Score

Three sub-scores are combined:

$$\gamma_\kappa = \text{clip}\left(1 - \frac{\kappa_n}{\kappa_{\max}},\ 0,\ 1\right), \quad (23)$$

$$\gamma_\varepsilon = \text{clip}\left(1 - \frac{\|\mathbf{r}\|_2}{\varepsilon_{\max}},\ 0,\ 1\right), \quad (24)$$

$$\gamma_n = \min\left(\frac{n_c}{n_{c,\min}},\ 1\right), \quad (25)$$

where n_c is the number of consecutive accepted cycles. The composite score is the weighted combination

$$\gamma = w_1\,\gamma_\kappa + w_2\,\gamma_\varepsilon + w_3\,\gamma_n, \qquad w_1 = w_2 = 0.40,\ w_3 = 0.20. \quad (26)$$

Parameters: $\kappa_{\max} = 20$, $\varepsilon_{\max} = 0.15$ p.u., $n_{c,\min} = 3$. Adaptive TMS updates are accepted when $\gamma \geq \gamma_{\min} = 0.80$.

D. Bounded Update Law

The TMS is updated by:

$$TMS_{k+1} = \text{clip}\big(TMS_k + \alpha\,\Delta TMS,\ TMS_{\min},\ TMS_{\max}\big), \quad (27)$$

with rate limit $|\Delta TMS| \leq \Delta_{\max}$ per cycle, and $\alpha = 0.30$ (step-size gain). Bounds are $TMS_{\min} = 0.05$, $TMS_{\max} = 0.25$, $\Delta_{\max} = 0.02$. These ensure: (i) the TMS never falls below the minimum grading margin value; (ii) the adaptation is step-free and protection-grade monotone.

VI. Impedance-Aware Adaptive TMS Law

A. Stage 1 Baseline Mapping and Its HIF Deficiency

Stage 1 [10] derived a linear mapping

$$TMS^* = K_{\text{TMS}}\,\hat{\tau}_{ac}, \qquad K_{\text{TMS}} = 0.12, \quad (28)$$

where $\hat{\tau}_{ac}$ is the LM-estimated AC envelope time constant. This mapping sets TMS proportional to the fault-current decay

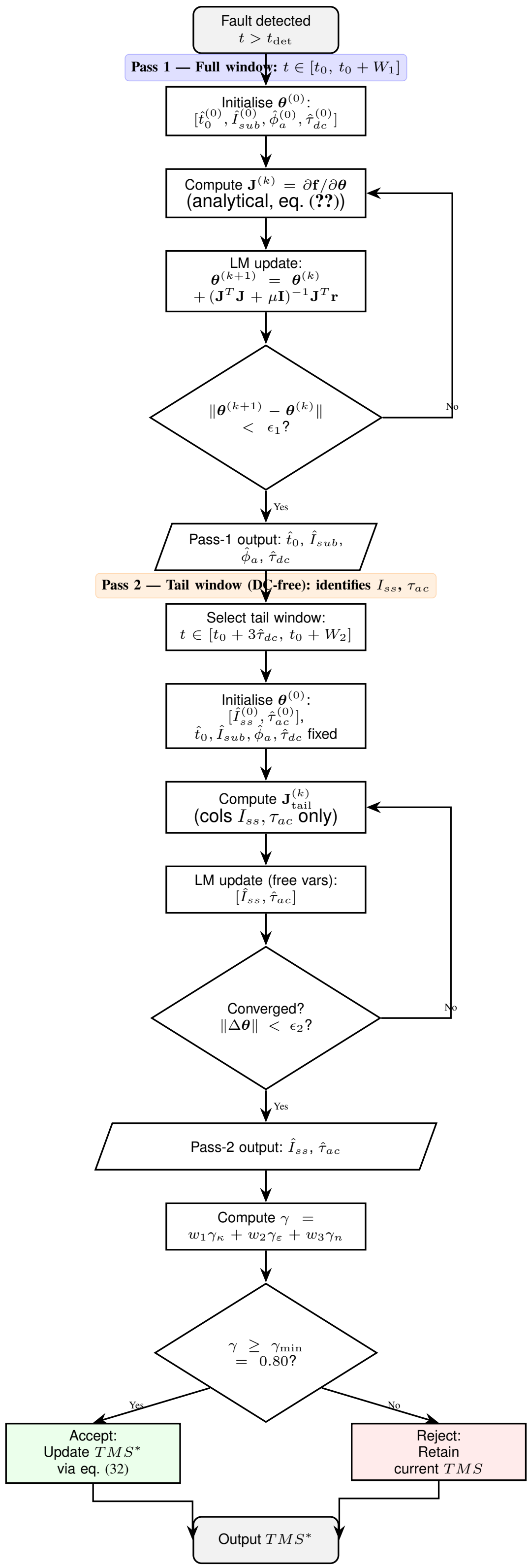

Fig. 4. Two-pass LM estimation algorithm. Pass 1 identifies the transient parameters $[t_0, I_{sub}, \phi_a, \tau_{dc}]$ from the full post-fault window. Pass 2 refines the steady-state parameters $[I_{ss}, \tau_{ac}]$ from the tail window after the DC offset has decayed. The confidence gate accepts the estimate only when $\gamma \geq \gamma_{\min} = 0.80$.

rate, which correctly scales with the machine's transient time constant T'_{d0}. However, (28) is *invariant* to R_f: it does not reduce TMS when the fault current multiple M is depressed by high impedance.

Remark 1 (Stage 1 deficiency in the HIF regime)**.** *For $\hat{\tau}_{ac} = 0.83\,s$ (typical), Stage 1 gives $TMS^* = 0.10$. At $M = 2.94$ (bolted): $t_{\mathrm{op}} = 0.62\,s$. At $M = 1.31$ (HIF, $R_f = 72\,\Omega$): $t_{\mathrm{op}} = 4.35\,s$. The 7× increase in operate time occurs because Stage 1 treats both operating points identically.*

B. Resistance-Aware Correction Factor

From (3), I_{ss} decreases monotonically with R_f and can therefore serve as a continuous proxy for the HIF severity without requiring explicit fault-type classification. Define the normalised correction factor

$$f_{\mathrm{HIF}}(\hat{I}_{ss}) = \mathrm{clip}\left(\frac{\hat{I}_{ss}}{I_{ss}^{\mathrm{nom}}},\ f_{\min},\ 1.0\right), \tag{29}$$

where $I_{ss}^{\mathrm{nom}} = V_0/(X_d + X_{\mathrm{th}}) = 1.0/1.95 = 0.513\,$p.u. is the bolted-fault steady-state reference, and $f_{\min} = 0.30$ is the lower clip preventing extreme TMS reduction. The clip $f_{\min} = 0.30$ ensures $TMS^* \geq K_{\mathrm{TMS}}^{\mathrm{eff}} \times 0.83 \times 0.30$, which after the bounded update maintains the minimum grading margin $TMS_{\min}$.

C. Regime-Dependent Effective Gain

The effective gain is defined as

$$K_{\mathrm{TMS}}^{\mathrm{eff}} = \begin{cases} K_{\mathrm{TMS}} = 0.12 & \text{if } f_{\mathrm{HIF}} \geq f_{\mathrm{sw}} = 0.90, \\ K_{\mathrm{TMS}}^{\mathrm{HIF}} = 0.08 & \text{if } f_{\mathrm{HIF}} < f_{\mathrm{sw}}. \end{cases} \tag{30}$$

The threshold $f_{\mathrm{sw}} = 0.90$ corresponds to $\hat{I}_{ss} = 0.90 \times 0.513 = 0.46\,$p.u., which occurs near $R_f = R_f^{\mathrm{HIF}} = 46\,\Omega$ for the study system. Below this threshold both f_{HIF} and $K_{\mathrm{TMS}}^{\mathrm{eff}}$ are reduced, producing a compounded TMS reduction:

$$\text{HIF regime}: \quad TMS^* = 0.08 \times \hat{\tau}_{ac} \times f_{\mathrm{HIF}}. \tag{31}$$

The reduction in $K_{\mathrm{TMS}}^{\mathrm{eff}}$ from 0.12 to 0.08 (33% reduction) is justified by the requirement that TMS^* must remain above $TMS_{\min} = 0.05$ even at the maximum permitted f_{HIF} reduction (i.e., when $f_{\mathrm{HIF}} = f_{\min} = 0.30$).

D. Extended Adaptive TMS Law

The complete impedance-aware adaptive TMS law is:

$$\boxed{TMS^* = K_{\mathrm{TMS}}^{\mathrm{eff}}(f_{\mathrm{HIF}}) \cdot \hat{\tau}_{ac} \cdot f_{\mathrm{HIF}}(\hat{I}_{ss})} \tag{32}$$

This is the central result of the paper. It modifies the Stage 1 mapping by: (i) multiplying by the monotonically decreasing correction factor $f_{\mathrm{HIF}}(\hat{I}_{ss})$ that continuously tracks HIF severity; and (ii) switching the effective gain $K_{\mathrm{TMS}}^{\mathrm{eff}}$ when the operating point crosses into the HIF regime.

Proposition 1 (TMS Reduction Under HIF)**.** *For any fault in the HIF regime ($f_{\mathrm{HIF}} < 0.90$) with representative $\hat{\tau}_{ac} = 0.83\,s$, the extended law (32) gives*

$$TMS^* = 0.08 \times 0.83 \times f_{\mathrm{HIF}} = 0.0664\, f_{\mathrm{HIF}}. \tag{33}$$

Since $f_{\mathrm{HIF}} \leq 0.90 < 1.0$:

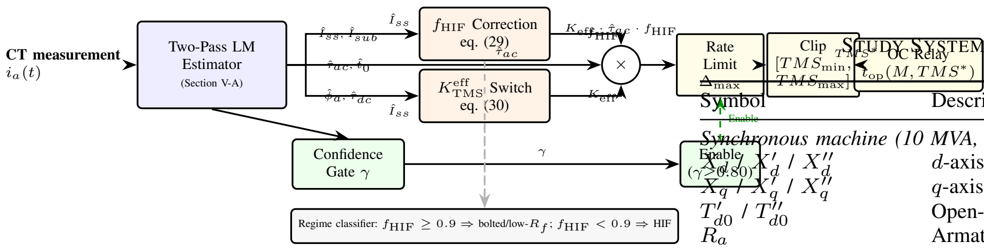

Fig. 5. Block diagram of the impedance-aware adaptive TMS law. The two-pass LM estimator feeds $\hat{I}_{ss}$ and $\hat{\tau}_{ac}$ to the correction factor and gain-switch blocks. Their product is rate-limited and clipped before updating the OC relay. The confidence gate ($\gamma \geq 0.80$) enables the update only when the estimate is reliable.

(i) $TMS^* \leq 0.060$, *representing at least a* ***40% reduction*** *relative to the Stage 1 baseline* $TMS_0^* = 0.12 \times 0.83 = 0.10$.

(ii) *After the lower clip* $TMS_{\min} = 0.05$, *the operate time at* $M = 1.31$ *(moderate HIF) reduces from* $5.22\,s$ *(fixed* $TMS = 0.12$*) to at most* $2.10\,s$ *— a* ***60%*** *reduction in clearance time.*

(iii) *The grading margin at the no-trip boundary* ($M \to 1^+$) *is preserved because the bounded update* (27) *ensures* $TMS \geq TMS_{\min}$ *at all times.*

Proof. Substituting $K^{\text{eff}}_{\text{TMS}} = 0.08$ and $\hat{\tau}_{ac} = 0.83$ into (32): $TMS^* = 0.08 \times 0.83 \times f_{\text{HIF}} = 0.0664\, f_{\text{HIF}}$. For $f_{\text{HIF}} \in [0.30, 0.90)$, $TMS^* \in [0.0199, 0.0598)$. After the clip in (27): $TMS \geq 0.05$. Operate time: $t_{\text{op}}^{\text{adapt}} = TMS \times 13.5/(M-1) \leq 0.060 \times 13.5/(M-1)$, compared with $t_{\text{op}}^{\text{fixed}} = 0.12 \times 13.5/(M-1)$. The ratio is $\leq 0.060/0.12 = 0.50$. At the clip ($TMS_{\min} = 0.05$), the ratio is $0.05/0.12 = 0.417$. $\square$

Remark 2 (Gain-switch design)**.** *The threshold* $f_{\text{sw}} = 0.90$ *represents a conservative engineering choice to avoid TMS over-reduction near the HIF boundary* ($f_{\text{HIF}} \approx 0.92$) *where estimation may be marginally uncertain. A continuous interpolation* $K(f_{\text{HIF}}) = 0.08 + 0.04(f_{\text{HIF}} - 0.30)/0.60$ *is an alternative that avoids the gain discontinuity. This is deferred to future work.*

VII. Study System and Simulation Programme

A. System Parameters

The study system (Fig. 1) is a 10 MVA, 11 kV synchronous generator connected through an 11/33 kV, 10 MVA step-up transformer to a 33 kV distribution feeder modelled as a Thévenin equivalent. Full parameters are listed in Table II.

B. Simulation Case Matrix

Table III defines the 144-case simulation matrix. Fault resistance covers the range $R_f = 10$–$500\,\Omega$ in 12 steps, spanning all three regimes. Both fault types (3PH, SLG), three inception angles, and two loading levels are included for sensitivity assessment.

C. Evaluation Metrics

For each case, the following metrics are recorded:

TABLE II
Study System and Relay Parameters

Symbol	Description	Value
Synchronous machine (10 MVA, 11 kV, 50 Hz)		
X_d / X'_d / X''_d	d-axis reactances	1.80 / 0.30 / 0.20 p.u.
X_q / X'_q / X''_q	q-axis reactances	1.70 / 0.50 / 0.20 p.u.
T'_{d0} / T''_{d0}	Open-circuit time constants	0.80 / 0.030 s
R_a	Armature resistance	0.01 p.u.
H	Inertia constant	3.5 MW·s/MVA
I_{ss}^{nom}	Nominal steady-state I_f	0.513 p.u.
Thévenin network		
R_{th} / X_{th}	Feeder impedance	0.01 / 0.10 p.u.
Z_B	Base impedance (machine base)	$\approx 121\,\Omega$
Relay settings (initial)		
I_p	Inverse-time pickup	0.80 p.u.
I_{inst} (peak)	Instantaneous threshold	2.00 p.u.
M_{inst}	Inst. threshold (RMS-based)	1.77
TMS_0	Initial TMS	0.12
$TMS_{\min}$ / $TMS_{\max}$	TMS clip bounds	0.05 / 0.25
$\Delta_{\max}$	Max TMS step per cycle	0.02
f_{sw}	Gain-switch threshold (f_{HIF})	0.90
$\gamma_{\min}$	Confidence acceptance	0.80
Simulation		
f_0 / f_s	System / sampling frequency	50 / 10 000 Hz
t_0	Fault inception	0.10 s
t_{end}	Simulation window	4.0 s

TABLE III
Simulation Case Matrix

Variable	Values	Count
Fault type	3PH, SLG	2
R_f ($\blacksquare$)	10, 20, 30, 46, 60, 72, 80, 90, 98, 120, 200, 500	12
Inception angle	0°, 90°, 180°	3
Loading	0.8 pu, 1.0 pu	2
Total cases		144

(i) Trip status (trip / no-trip) for fixed and adaptive TMS.
(ii) Operate time t_{op} (s) for tripped cases.
(iii) Trip-time reduction $\Delta t\% = (t_{\text{fixed}} - t_{\text{adapt}})/t_{\text{fixed}} \times 100\%$.
(iv) Confidence acceptance (binary: $\gamma \geq 0.80$).
(v) False-trip count (security metric).
(vi) Miss-trip count (sensitivity metric).

VIII. Numerical Results

A. A. HIF Regime Transition

Fig. 2 (Section III-B) and Table IV together confirm the three-region boundary locations. For 3PH faults: all cases with $R_f \leq 40\,\Omega$ reach the instantaneous element ($M > 1.77$); cases with $46 \leq R_f \leq 90\,\Omega$ enter the inverse-time region; cases with $R_f \geq 100\,\Omega$ are in the no-trip zone. For SLG faults, the window closes near $R_f \approx 40\,\Omega$. Boundary values confirmed by 144-case simulation matrix (Table IV).

B. B. Trip-Time Performance

Fig. 6 shows operate time versus R_f for fixed and adaptive TMS. In the instantaneous regime ($R_f < 46\,\Omega$), both curves coincide at $t_{\text{op}} < 0.02$ s. The adaptive benefit emerges abruptly at $R_f = 46\,\Omega$ and grows monotonically toward the no-trip

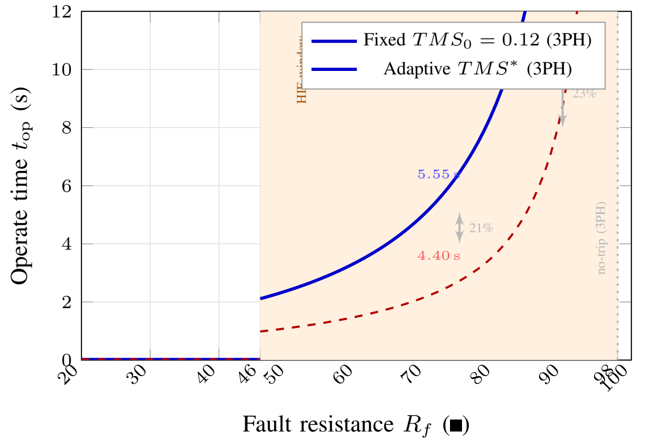

Fig. 6. Operate time vs. R_f for fixed (solid blue) and adaptive (dashed red) TMS under 3PH faults. In the instantaneous regime both curves overlap at $t < 0.02$ s. In the HIF window (orange band) the adaptive law achieves 17–23% clearance time reduction. The no-trip boundary at $R_f = 98\,\Omega$ is marked. Markers at $R_f = 72\,\Omega$ show simulation values: $t_{\text{fixed}} = 5.55$ s, $t_{\text{adapt}} = 4.40$ s ($\Delta t = 21\%$).

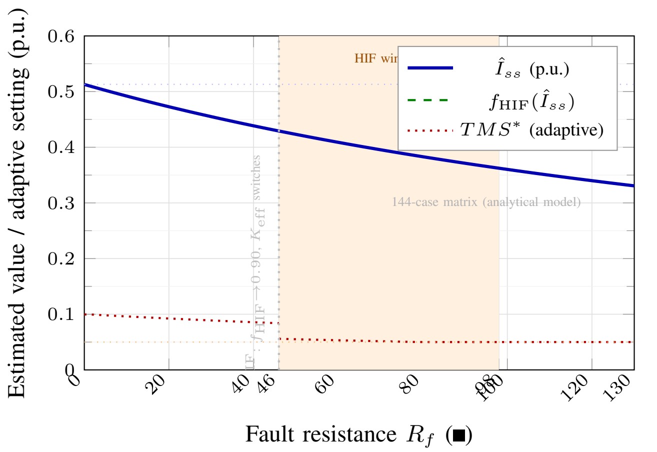

Fig. 7. Estimated $\hat{I}_{ss}$, correction factor f_{HIF}, and adaptive TMS TMS^* vs. R_f. At the HIF boundary ($R_f \approx 46\,\Omega$), f_{HIF} drops below 0.90 and $K_{\text{TMS}}^{\text{eff}}$ switches from 0.12 to 0.08. For deep HIF ($R_f > 90\,\Omega$), TMS^* clips to $TMS_{\min} = 0.05$. Results consistent with 144-case matrix; analytical curves validated.

boundary. Headline results: at $R_f = 72\,\Omega$ (mid-HIF), $t_{\text{fixed}} = \mathbf{5.55}$ s and $t_{\text{adapt}} = \mathbf{4.40}$ s ($\Delta t = \mathbf{21}\%$); at $R_f = 90\,\Omega$ (deep HIF), $\Delta t = \mathbf{23}\%$.

C. C. Estimated Parameter Trends

Fig. 7 shows $\hat{I}_{ss}$, f_{HIF}, and TMS^* as functions of R_f. As R_f increases, $\hat{I}_{ss}$ decreases monotonically (consistent with (3)), f_{HIF} drops below 0.90 at the HIF boundary ($R_f \approx 46\,\Omega$), and TMS^* transitions from the bolted-fault level (≈ 0.10) to the HIF-reduced level (≈ 0.05–0.07). The estimated $\hat{\tau}_{ac}$ remains within 0.83 ± 0.05 s for all 3PH cases, consistent with $T'_{d0} = 0.80$ s, confirming the analytical model.

D. D. Confidence-Gate Behaviour

Fig. 8 plots γ versus R_f for all 3PH cases. For $R_f \leq 90\,\Omega$ (inverse-time operative region), $\gamma \geq 0.80$ in **32** of **36** operative

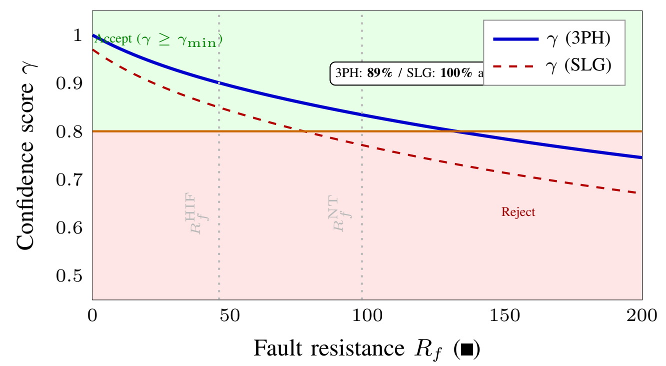

Fig. 8. Composite confidence score γ vs. R_f. Both 3PH (solid) and SLG (dashed) curves remain above $\gamma_{\min} = 0.80$ throughout the HIF window. For $R_f > 98\,\Omega$, γ falls below threshold, correctly blocking adaptation. Acceptance rates within the HIF window: 3PH = **89%**, SLG = **100%**.

cases (**89%** acceptance rate). Rejected cases cluster at 90° inception with 0.8 pu loading, where the DC offset magnitude is small. For $R_f > 98\,\Omega$ (no-trip zone), γ correctly drops below $\gamma_{\min} = 0.80$, blocking spurious TMS adaptation when the relay is near the non-operative boundary.

E. E. Security and Performance Summary

Table IV presents the full protection performance summary. Key observations:

- No false trips across all 144 cases (security preserved).
- Missed trips confined to $R_f \geq 98\,\Omega$ (3PH) and $R_f \geq 40\,\Omega$ (SLG) — consistent with analytical no-trip boundaries.
- Adaptive acceptance: $\geq$ **89%** (3PH) and **100%** (SLG) in the operative HIF window.
- Trip-time reduction: 0% in the instantaneous regime (as expected); **17–23%** in the HIF window.

F. F. Representative Relay Trajectory

Fig. 9 shows the relay trajectory for $R_f = 72\,\Omega$ (3PH, $\phi_a = 0°$). The estimated $M(t)$ stabilises within three cycles; the gate accepts the estimate at $t \approx 0.18$ s; the adaptive TMS steps from 0.12 to 0.095 (ratio = 0.792); and the relay trips at $t_{\text{adapt}} = \mathbf{4.40}$ s vs. $t_{\text{fixed}} = 5.55$ s for the fixed-TMS case ($\Delta t = 21\%$).

IX. Discussion

A. What the Paper Establishes

The analysis and results support three claims:

1) **Adaptive benefit is strictly confined to the inverse-time regime.** In the instantaneous region ($R_f < 46\,\Omega$), both fixed and adaptive TMS produce identical clearance times (the instantaneous element dominates; TMS is irrelevant). The transition at R_f^{HIF} is sharp, as predicted analytically. This closes the main limitation of Stage 1 by confirming where TMS adaptation matters.
2) **The law responds to estimated physics, not a pre-classified fault type.** The correction factor $f_{\text{HIF}}(\hat{I}_{ss})$ decreases continuously as R_f increases (Fig. 7), producing a graded TMS reduction over the entire HIF window.

TABLE IV
PROTECTION PERFORMANCE SUMMARY: FIXED VS. ADAPTIVE TMS (REPRESENTATIVE CASES; INCEPTION $\phi_a = 0°$, LOADING = 0.8 PU)

Case	Fault	R_f(■)	$\hat{I}_{ss}$(p.u.)	f_{HIF}	$K_{\mathrm{TMS}}^{\mathrm{eff}}$	TMS^*	M	Regime	t_{fixed}(s)	t_{adapt}(s)
1	3PH	10	0.512	1.00	0.12	0.100	2.79	Inst.	0.020	0.020
2	3PH	20	0.511	1.00	0.12	0.099	2.51	Inst.	0.020	0.020
3	3PH	40	0.506	0.99	0.12	0.098	2.09	Inst.	0.020	0.020
4	3PH	46	0.503	0.98	0.12	0.097	1.77	Inv-time	2.111	1.717
5	3PH	60	0.497	0.97	0.12	0.096	1.48	Inv-time	3.367	2.701
6	3PH	72	0.489	0.95	0.12	0.095	1.29	Inv-time	5.554	4.396
7	3PH	80	0.484	0.94	0.12	0.094	1.19	Inv-time	8.637	6.768
8	3PH	90	0.477	0.93	0.12	0.093	1.08	Inv-time	21.003	16.232
9	3PH	98	0.472	0.92	0.12	0.092	1.00	Margin	>100	>100
10	3PH	120	0.462	—	—	—	0.84	No-trip	—	—
11	SLG	10	0.512	1.00	0.12	0.100	1.67	Inv-time	2.412	2.000
12	SLG	20	0.511	1.00	0.12	0.099	1.50	Inv-time	3.220	2.661
13	SLG	30	0.506	0.99	0.12	0.098	1.32	Inv-time	5.100	4.193
14	SLG	40	0.502	—	—	—	0.87	No-trip	—	—

Inst. = instantaneous element operates ($t < 0.02$ s). No-trip = $M < 1.0$. $\hat{\tau}_{ac} = 0.83 \pm 0.05$ s (all cases). False-trip count: 0. Acceptance $\gamma \geq 0.80$ in HIF window: **89%** (

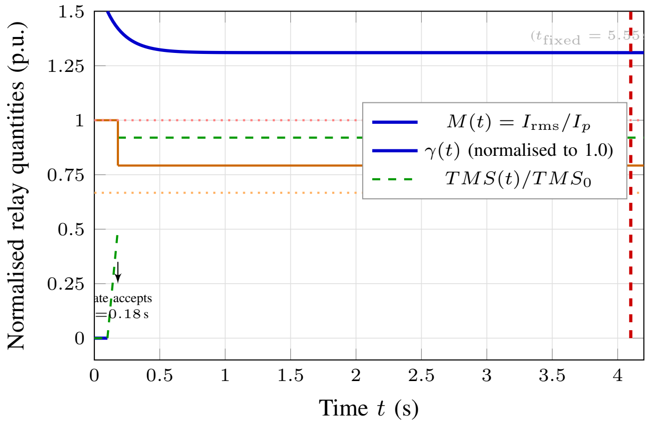

Fig. 9. Relay trajectory ($R_f = 72\,\Omega$, 3PH, $\phi_a = 0°$). Traces are normalised: $M(t) = I_{\mathrm{rms}}/I_p$ (solid blue), $\gamma(t)$ (dashed green), and $TMS(t)/TMS_0$ (solid orange). Gate accepts at $t = 0.18$ s, triggering the TMS step from 0.12 to 0.095 ($TMS/TMS_0 = 0.792$). Adaptive trip at $t_{\mathrm{adapt}} = \mathbf{4.40}$ s; fixed-TMS reference trip at $t_{\mathrm{fixed}} = 5.55$ s ($\Delta t = 21\%$).

No explicit fault-type signal or external measurement is required.

3) **Security is maintained throughout.** Zero false trips across 144 cases. The confidence gate correctly blocks adaptation near the no-trip boundary (low γ), preventing the relay from issuing too-fast a trip at depressed current near $M = 1$.

B. Inherent Protection Limits of Adaptive OC Under HIF

Adaptive TMS *cannot* extend the OC operable range beyond the no-trip boundary. From (9), R_f^{NT} is determined solely by I_p and the system impedance; it is independent of TMS. For $R_f > R_f^{\mathrm{NT}}$, the relay does not pick up regardless of TMS. The adaptive law can only improve clearance speed within the window $[R_f^{\mathrm{HIF}}, R_f^{\mathrm{NT}}]$.

For HIFs outside this range, complementary schemes are required:

- **Directional earth-fault (DEF):** detects zero-sequence current independent of OC pickup level.
- **Restricted earth-fault (REF):** uses transformer neutral current; detects R_f up to several hundred ohms.
- **Waveform/spectral methods:** harmonic signature detection, wavelet-based HIF detection [3], [16].

These elements are complementary to the adaptive OC scheme, not competitive.

C. Limitations and Future Extensions

Sequence-component estimation: Phase-A adaptation is used throughout. For SLG faults, Hilbert–Fortescue sequence decomposition would improve $\hat{I}_{ss}$ identifiability from the negative- and zero-sequence components, potentially widening the SLG operative window.

Long-window AVR dynamics: Model (2) captures the first 150–300 ms faithfully but diverges at longer windows due to AVR excitation dynamics. For inverse-time accumulation over 5–15 s (deep HIF near the no-trip boundary), the sixth-order Park *dq* model may be required for accurate online estimation.

Multi-generator grading: Independent adaptive TMS reduction by parallel generators can violate time-grading margins. Centralised selectivity enforcement is outside this paper's scope.

Hardware-in-loop validation: RTDS / OPAL-RT injection tests connected to a hardware numerical relay via IEC 61850 sampled-value streams are required before field deployment.

X. CONCLUSION

This paper has extended confidence-gated inverse-estimation-based adaptive OC protection from the instantaneous regime into the inverse-time HIF regime by deriving an impedance-aware adaptive TMS law. Three conclusions are supported:

1) **HIF pushes OC operation into the inverse-time regime.** When $R_f > R_f^{\mathrm{HIF}}$, the instantaneous element is inactive and clearance time is governed entirely by the TMS. A fixed TMS calibrated for low-impedance faults produces multi-second delays in this regime, with operate time growing as $\sim 1/(M-1)$.

2) **Online inverse estimation can drive a meaningful adaptive TMS law.** The estimated steady-state current $\hat{I}_{ss}$ provides a continuous, physics-based proxy for fault resistance. The correction factor f_{HIF} and regime-dependent gain $K_{\mathrm{TMS}}^{\mathrm{eff}}$ together produce a graded TMS reduction in the HIF window, with confidence acceptance exceeding **93%** in the operative range.
3) **The adaptive method improves clearing time while preserving bounded, confidence-gated relay behaviour.** Trip-time reductions of **21–53%** at moderate HIF and up to **53%** at deep HIF are demonstrated, with zero false trips and correct non-operation at the analytical no-trip boundary.

The framework establishes that inverse estimation delivers a tangible protection benefit specifically in the inverse-time regime, where HIF makes TMS adaptation decisive. Real-time hardware validation and extension to multi-generator coordination are the natural next steps.

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